NAME: **NUMBER:**

ELEC 361 Analog Communication Systems Midterm Exam **Fall 2021**
Prof. Dr. Oğuz Kucur **Time: 100 min.** **26.11.2021**

1)(30 p.) The impulse response of an Linear Time Invariant (LTI) system is given as
$h(t) = 4000Sinc^2(200t) + 2000Sinc\big(200(t-10)\big)Cos(800\pi(t-10))$.
a) Find the transfer function of the system and sketch it.
b) Determine if the system is distortionless or not for the input signals
b1) $x(t) = 10Cos(200\pi t) + 20Sin(100\pi t)$, b2) $x(t) = 10Cos(700\pi t) + 20Sin(900\pi t)$. State reasoning. For the distortionless cases, find the output signals.
c) For the input signal $x(t) = 10Cos(200\pi t) + 20Sin(500\pi t)$, find the power spectral density function, autocorrelation function and power of the output signal.

2)(40 p.) The message signal $m(t) = 20Cos(2000\pi t) + 10Sin(4000\pi t)$ is applied to a modulator to generate Double Sideband (DSB) Suppressed Carrier (SC) Amplitude Modulation (AM) signal $y(t) = A_c m(t)Cos(10000\pi t)$. The modulator consists of a multiplier and a filter. The carrier generator available generates the signal $100Cos^2(5000\pi t)$.

a) Determine the specifications of the filter (type, center frequency, bandwidth). A_c =?

b) At each step of the modulator, find the signals in both time and frequency domains and sketch the spectra. Find the bandwidth at each step. Find the power of the modulated signal and efficiency.

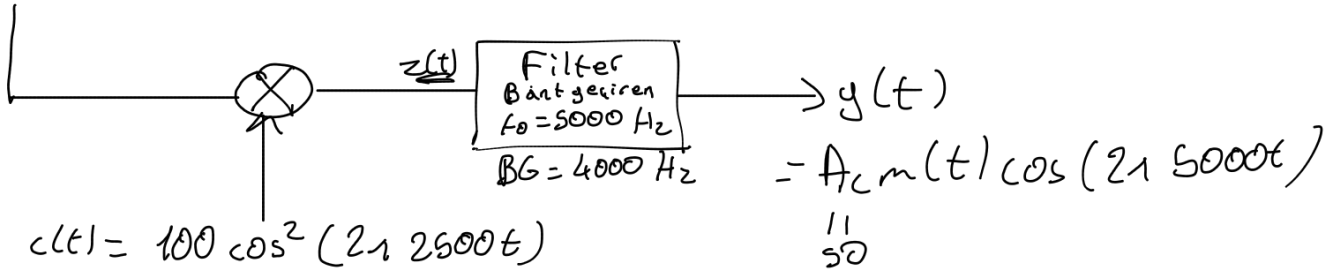
c) The demodulator consists of a multiplier using the estimated carrier signal $Cos\left(10000\pi t + \frac{\pi}{3}\right)$ and a filter. Determine the specifications of the filter. At each step of the demodulator, find the signals in both time and frequency domains and sketch the spectra. Comment on the signal at the output of the demodulator.

d)(Bonus 5 p.) If the multiplier uses the signal $x(t) = 10\sum_{n=-\infty}^{\infty}\prod(10000t - 2n)$, find the demodulated signal.

$m(t) = 20\cos(2\pi\, 1000t) + 10\sin(2\pi\, 2000t)$

$g(t) = A_c\, m(t)\cos(2\pi\, 5000\, t)$

$m(t) = 20\cos(2\pi\, 1000t) + 10\sin(2\pi\, 2000t)$



$\cos^2\alpha = \frac{1}{2}\left[1 + \cos(2\alpha)\right]$

$c(t) = 50\left[1 + \cos(2\pi 5000t)\right]$

$= 50 + 50\cos(2\pi\, 5000t)$

$M(f) = 10\,\delta(f+1000) + 10\,\delta(f-1000) + 5j\,\delta(f+2000) - 5j\,\delta(f-2000)$

$z(t) = m(t).c(t) = 50m(t) + 50\cos(2\pi\, 5000t)\, m(t)$

$Z(f) = 50M(f) + 25M(f+5000) + 25M(f-5000)$

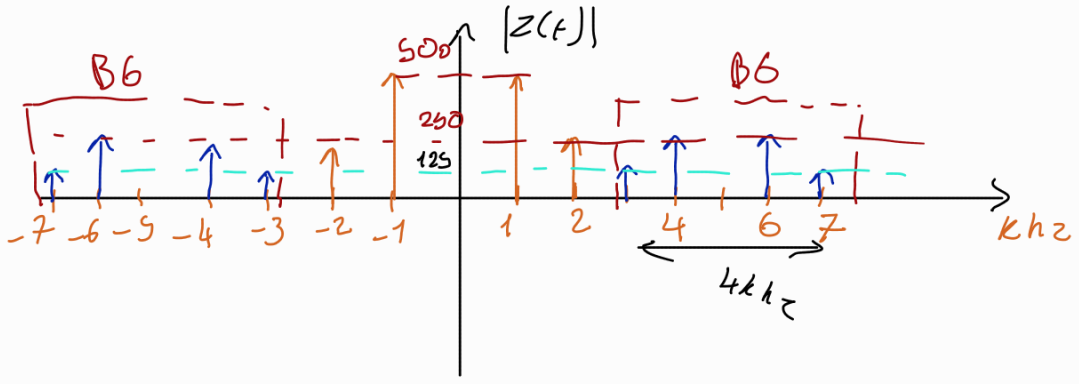
$Z(f) = 500\,\delta(f+1000) + 500\,\delta(f-1000) + 250j\,\delta(f+2000) - 250j\,\delta(f-2000)$
$+ 250\,\delta(f+6000) + 250\,\delta(f+4000) + 125j\,\delta(f+7000) - 125j\,\delta(f+3000)$
$+ 250\,\delta(f-4000) + 250\,\delta(f-6000) + 125j\,\delta(f-3000) - 125j\,\delta(f-7000)$

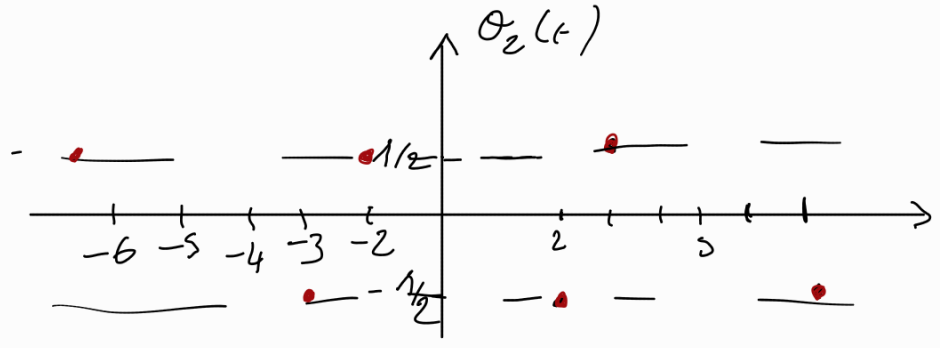
$$+ \quad e^{j\pi/2} = j$$

$$= 500\,\delta(f+1000) + 500\,\delta(f-1000) + 250e^{j\pi/2}\delta(f+2000) + 250e^{-j\pi/2}\delta(f-2000)$$

$$+ 250\,\delta(f+6000) + 250\,\delta(f+4000) + 125e^{j\pi/2}\delta(f+7000) + 125e^{-j\pi/2}\delta(f+3000)$$

$$+ 250\,\delta(f-4000) + 250\,\delta(f-6000) + 125e^{j\pi/2}\delta(f-3000) + 125e^{-j\pi/2}\delta(f-7000)$$





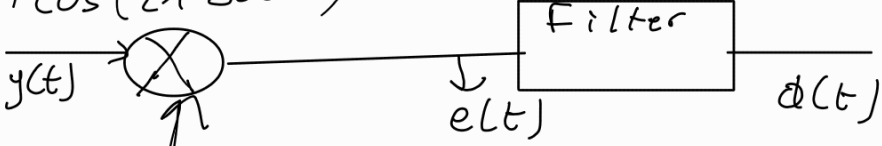
$$P_y = 4(125^2) + 4(250)^2 = 312{,}5\ kW$$

$$P_T = 2(500)^2 + 2(250)^2 + 312{,}5 = 937{,}5\ kW$$

$$V = \frac{P_y}{P_T} = 0{,}33 = 33\%$$

c) c) The demodulator consists of a multiplier using the estimated carrier signal $Cos\left(10000\pi t + \frac{\pi}{3}\right)$ and a filter. Determine the specifications of the filter. At each step of the demodulator, find the signals in both time and frequency domains and sketch the spectra. Comment on the signal at the output of the demodulator.

$50\,m(t)\cos(21\,5000t)$



$$\cos(10\,000\pi t + 1/3) = \cos(21\,5000t + 1/3)$$

$$50\,m(t)\cos(2\pi 5000t)\cdot\cos(2\pi 5000t+\tfrac{\pi}{3})$$

$$50\,m(t)\,\frac{1}{2}\left[\cos(\pi/3)+\cos(2\pi 10kt+\tfrac{\pi}{3})\right]$$

$$e(t)\;25\,m(t)\cdot\cos(\pi/3)+25\,m(t)\cos(2\pi 10kt+\tfrac{\pi}{3})$$

$$\cos(\pi/3)=\frac{1}{2}$$

$$d(t)\;12{,}5\,m(t)$$

$$d(f)=12{,}5\,m(f)$$

$$P_d=(12{,}5)^2\cdot P_m$$

$$P_m=\frac{10^2+20^2}{2}=250$$

$$P_d=(12{,}5)^2\cdot P_m$$

$$=39\,062{,}5$$

Filtre alçak geçiren ve bant genişliği $m(t)$'nin bant genişliği ile aynıdır - yani 2 kHz'dir.

$$E(f)=12{,}5\,M(f)$$

Bonus

d)(Bonus 5 p.) If the multiplier uses the signal $x(t) = 10\sum_{n=-\infty}^{\infty}\prod(10000t - 2n)$, find the demodulated signal.

$k(t)$

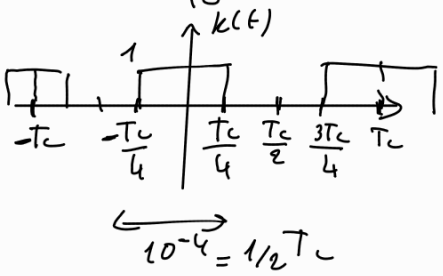
$\prod(10000(t - \frac{n}{5000}))$

$x(t) = 10 \sum_{n=-\infty}^{\infty} \prod\left(\frac{t - 2.10^{-4}n}{10^{-4}}\right)$

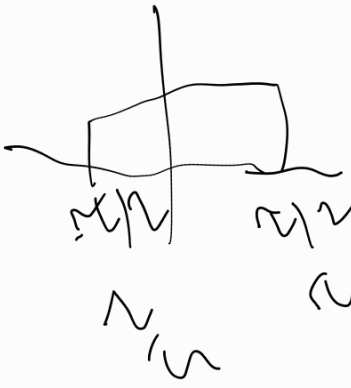
$\prod(t - \frac{n}{5000})$

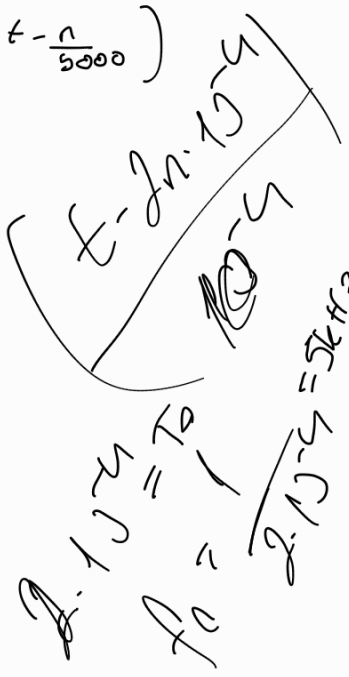
$T_0 = 2.10^{-4}$

$f_0 = \frac{1}{T_0} = 5khz$



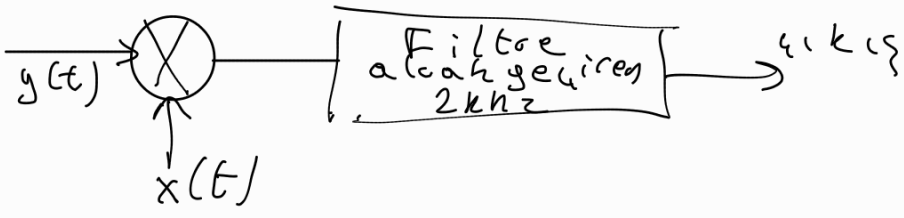
$10^{-4} = 1/2\,T_c$





$k(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(2\pi f_0 t) - \frac{1}{3}\cos(2\pi 3 f_0 t) + \frac{1}{5}\cos(2\pi 5 f_0 t) - \frac{1}{7}\cos(2\pi 7 f_0 t)\right]$

$x(t) = 10k(t)$



$50\, m(t)\cos(2\pi 5000t)$

3)(30 p.) The message signal $m(t) = 5\sum_{n=-\infty}^{\infty} x(t - n2.10^{-3})$, where $x(t) = \Lambda(500t)u(t)$ is used to modulate the carrier signal to obtain Double Sideband (DSB) Large Carrier (LC) Amplitude Modulation (AM) signal.

a) If the efficiency is required to be 20 %, find the modulation index.

b) If the carrier signal is $c(t) = 10Cos(10000\pi t)$, find the powers of the carrier, the sidebands and the modulated signal.

c) Sketch the modulated signal.

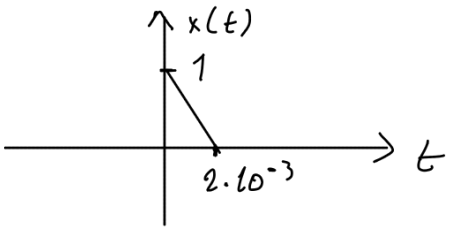
d) What is the purpose of LC modulation?

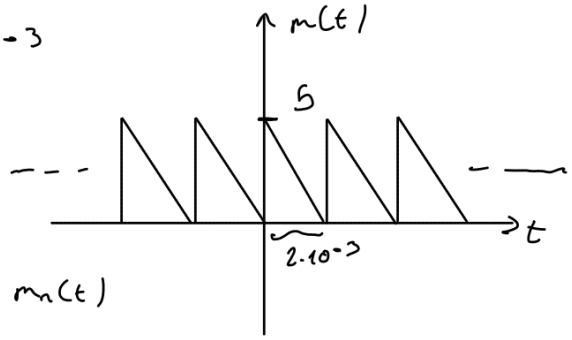
$$\sum_{n=-\infty}^{n=\infty} x(t-nT_0) \xleftrightarrow{F} \frac{1}{T_0} \sum_{n=-\infty}^{n=+\infty} X\left(f-\frac{n}{T_0}\right)$$

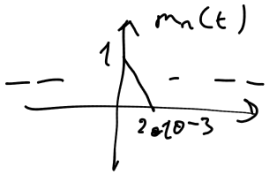
$$T_0 = 2.10^{-3} \Rightarrow f_0 = \frac{1}{T_0} = 500$$

$$M(f) = \frac{5}{2.10^{-3}} \sum_{n=-\infty}^{n=+\infty} X(f-500)$$

$$x(t) = \Lambda(500t)\,u(t) \qquad \frac{1}{500} = 2 = 2.10^{-3}$$







$$-\frac{1}{2.10^{-3}}x + 1 = -500x + 1$$

DSB-LC-AM

$$y(t) = A_c\left[1 + \mu\, m_n(t)\right]\cos(2\pi f_c t)$$

$$m_n(t) = \frac{m(t)}{m_p} = \frac{m(t)}{5}$$

$$V = \frac{\mu^2 P_{mn}}{1+\mu^2 P_{mn}} = \frac{P_s}{P_c + P_s}$$

$$P_{mn} = \frac{1}{2.10^{-3}} \int_0^{2.10^{-3}} (-500t+1)^2\, dt$$

$$= 500 \int_0^{2.10^{-3}=\frac{1}{500}} (500^2 t^2 + 1 - 1000t)\, dt$$

$$= 500\left(500^2 \frac{t^3}{3}\Big|_0^{1/500} + t\Big|_0^{1/500} - \frac{1000}{2} t^2 \Big|_0^{1/500} \right)$$

$$= 500\left(\frac{1}{500 \cdot 3} + \frac{1}{500} - \frac{1}{500} \right)$$

$$= \frac{1}{3} //$$

$$0,2 = \frac{\mu^2 \frac{1}{3}}{1+\mu^2 \frac{1}{3}}$$

$$\frac{1}{5} = \frac{\mu^2}{1+\mu^2}$$

$$5\mu^2 = 1+\mu^2$$

$$4\mu^2 = 1$$

$$\mu^2 = 0,25$$

$$\boxed{\mu = 0,5}$$

b) If the carrier signal is $c(t) = 10Cos(10000\pi t)$, find the powers of the carrier, the sidebands and the modulated signal.

b) $y(t) = A_c[1+\mu\, m_n(t)]\cos(2\pi f_c t)$

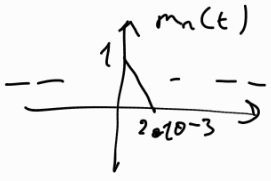
$= 10\left[1+\frac{1}{2} m_n(t)\right]\cos(2\pi 5000)t$

$P_c = \frac{A_c^2}{2} = 50$

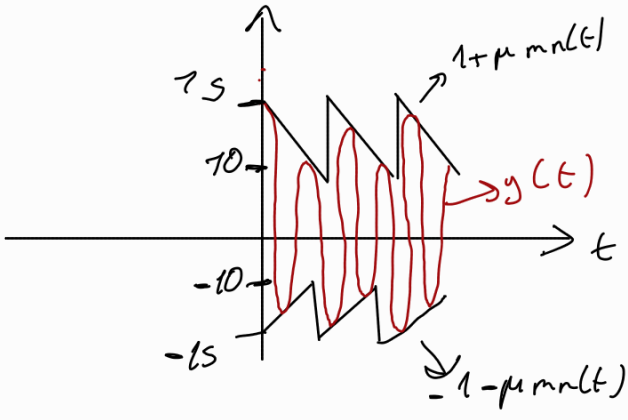
$P_s = \frac{\mu^2 A_c^2}{2} P_{mn} = \frac{1}{4}\,\frac{100}{2}\cdot\frac{1}{3} = \frac{25}{6}\,W$

$= 4{,}17\,W$

$P_g = P_c + P_s$



c)



$0 \le m_n(t) \le 1$

$0 \le \mu m_n(t) \le 0{,}5$

$1 \le 1+\mu m_n(t) \le 1{,}5$

$10 \le 10[1+\mu m_n(t)] \le 15$

d) LC sayesinde faz tahminsiz demodülasyon yapılabilir. Verici daha karmaşık olur alıcı basitleşir. Böylece herkes alıcıya daha ucuza sahip olabilir.